

Reza Shabrang Maryaan

Machine Learning Engineer, applying to Master's programmes in AI

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PROFILE

Machine learning engineer with 5+ years building production AI systems (NLP, retrieval-augmented generation, computer vision, graph methods, time series). Team lead of a four-person delivery group. Seeking an M1 in artificial intelligence to deepen theoretical foundations and move from applied engineering toward research, with a long-term interest in intelligent systems that act in the physical world.

EDUCATION

B.Sc. Industrial Engineering, Amirkabir University of Technology (Tehran Polytechnic) 2016 – 2021

- Overall average 16.77/20 (GPA 3.5/4); 3.7/4 over the last 56 units. Bachelor project 20/20, internship 20/20.
- Thesis: *Predicting construction project cost with machine learning* (supervisor Dr. Maryam Ashrafi). Decision trees, random forests, XGBoost and neural networks on 132 projects / 7,000+ data points; 20% improvement over traditional forecasting.
- Strong marks in linear algebra, probability, statistics, operations research, simulation and programming.

EXPERIENCE

Asam — Data Scientist → ML Software Engineer → Team Lead 2021 – present

- Built and operated 20+ production systems (AI products, NLP/ML services, data platforms) for media and fintech clients; lead a four-person delivery group.
- Designed a multi-step agent (LangGraph, RAG over isolated per-tenant knowledge bases) serving 30+ workspaces and tens of thousands of monthly conversations.
- Fine-tuned a Persian BERT model for offensive-comment classification (90% accuracy); built embedding-based text and image de-duplication with vector search (BGE-M3, Sentence-BERT, ResNet, VGG-16, Milvus).
- Standardised ML delivery with MLflow/Metaflow and reusable model-serving APIs on Kubernetes.
- Anomaly detection (Isolation Forest, forecasting with Darts) for a cryptocurrency exchange; ClickHouse analytics platform serving 100B+ records.

QuantRisk — Data Engineer 2020 – 2021

- Researched the US electricity market and designed ETL workflows for downstream analytics.

AhanOnline — BI Developer 2019 – 2020

- Dashboards and data integrations; ARIMA model for steel and iron price forecasting.

RESEARCH AND SELECTED PROJECTS

- **Publication:** co-author, *Automated LOINC Mapping of Persian-English Mixed-Language Clinical Laboratory Test Names*, Studies in Health Technology and Informatics, 2026 (DOI 10.3233/SHTI260091).
- **Computer vision:** implemented a Reconstructive Sequence-Graph Network for video summarisation; served image up-scaling models through an API.
- **Graph learning:** spectral clustering of 1M+ keywords (self-tuning variant); curated a GNN research resource (gnn-atlas) and study of GNN explainability with PyTorch Geometric.
- **COVID density forecasting** with Prophet, RNN and LSTM (5% MAPE); **optimisation** with genetic algorithms and TLBO.

TEACHING

- Instructor, Practical ML course (Asam, 2024) and Python programming course (AUT, 2023); teaching assistant for Data & Information Analysis (2019–20) and Management Information Systems (2018–19).

SKILLS AND LANGUAGES

ML: Python, PyTorch, TensorFlow, Transformers, RAG, LangGraph, scikit-learn, XGBoost, Darts, vector search.

Engineering: FastAPI, Node.js, Kubernetes, Docker, Kafka, ClickHouse, PostgreSQL, MLflow, GCP. **Maths:** linear algebra, probability, optimisation.

Languages: Persian (native) · English: Duolingo English Test 140 (CEFR C1, Jan 2025), TOEFL iBT 97 (2022) · French: beginner, currently learning.